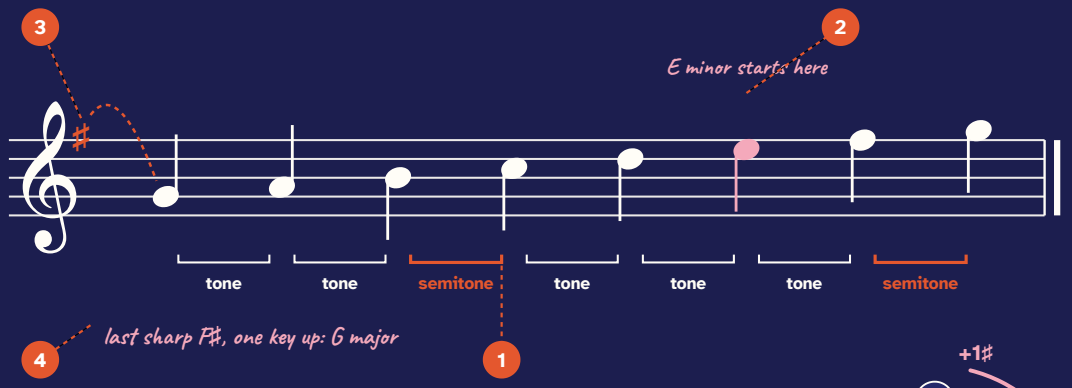
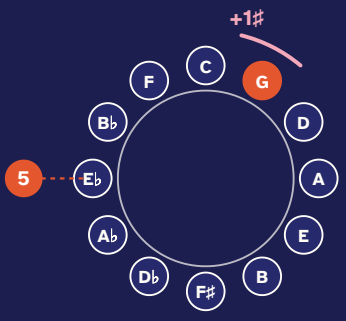
PART III · THE GROUND UNDER THE CHORDS

Scales and keys

Notes travel in families. Build the scales that whole pieces are made from, then learn to read the key a piece lives in: its signature, its home note, its place on the circle. All of it is in one scale.





1 **Major scales** p. 70
T T S T T T S: the two short steps

2 **Minor scales** p. 73
the same notes from the 6th note up

3 **Keys and key signatures** p. 76
the sharps at the start apply all the way

4 **Finding the key** p. 79
last sharp, one key up; second-last flat

5 **The circle of fifths** p. 82
clockwise, one more sharp each step

★ **Tips: when a note sounds wrong** p. 85

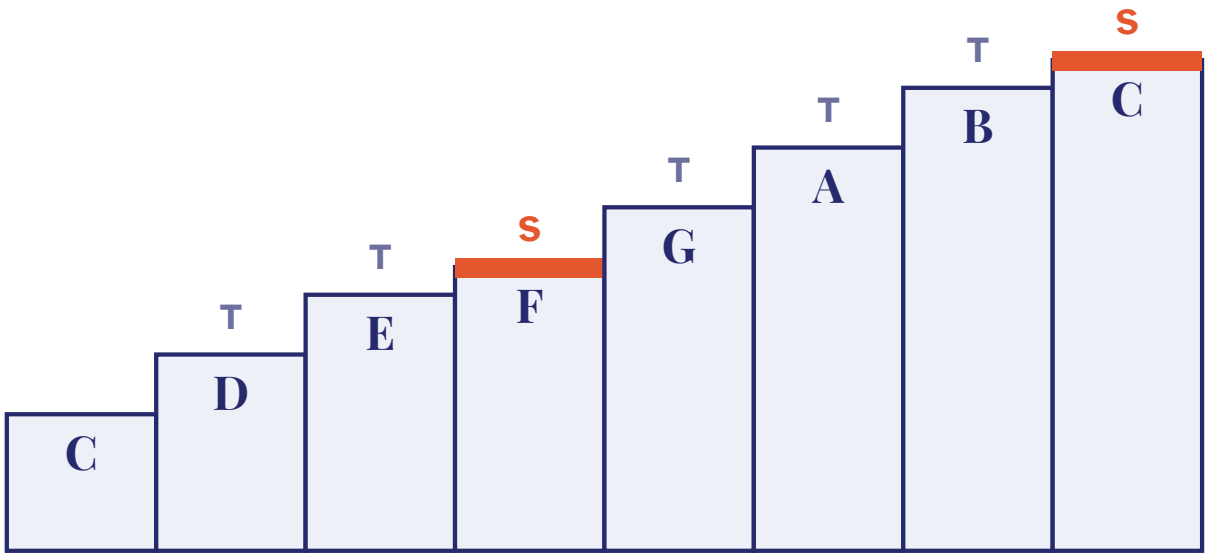
PART III · SCALES AND KEYS · 1

Major scales

Climb from a key note to the next one with the same name, following a fixed recipe of tones (T) and semitones (S): that is a **scale**. The recipe, not the starting note, makes it major or minor.

THE SECRET

Every major scale: tone, tone, semitone, tone, tone, tone, semitone.



Tone, tone, semitone, tone, tone, tone, semitone: the two short steps are 3–4 and 7–8

SCALE	STEPS	FIRST MET
major	T T S T T T S	Level ½
natural minor	T S T T S T T	Level ½
harmonic minor	T S T T S T+S S	Level ½
melodic minor (up)	T S T T T T S; down as the natural minor	Level 2
chromatic	every semitone	Level 2
pentatonic (black keys)	T T T+S T T+S	Level 0

Every step has a name

1 tonic · **2** supertonic · **3** mediant · **4** subdominant · **5** dominant · **6** submediant · **7** leading note

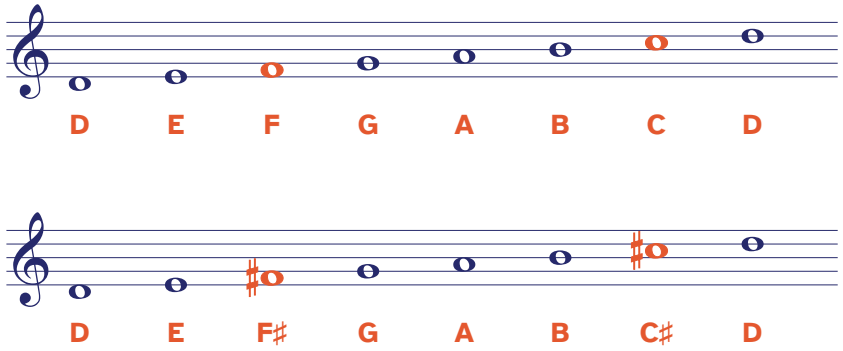
Major or minor?

The same key note can start a major or a minor scale: C major and C minor. The pattern decides. Major sounds bright and settled; minor, darker.

PART III · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 1

Build D major



1. **Write the eight letters** from D to D, every letter once.
2. **Check the steps** against T T S T T T S: E–F is too small, B–C is too small.
3. **Raise F and C** a semitone to fix them.

D major: **D E F♯ G A B C♯ D**, two sharps.

EXAMPLE 2

Build F major

1. **Write the letters** from F to F: F G A B C D E F.
2. **Check the steps** against T T S T T T S: A to B is a tone, but it must be a semitone.
3. **Lower B** to B♭: now A to B♭ is a semitone and B♭ to C a tone.

F major: **F G A B♭ C D E F**, one flat.

HOW TO

Practise a scale

1. **Hands separately,** slowly, saying the finger numbers aloud.
2. **Fingering for C, G, D, A and E major** (right hand): 1 2 3, thumb under, 1 2 3 4 5.
3. **Listen for evenness:** every note the same length and loudness.
4. **Then hands together,** one octave, then two.

GAME

Spot the wrong note

1

G A B C D E F G

2

D E F♯ G A B C D

3

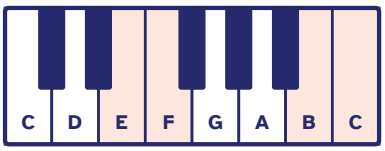
F G A B C D E F

Each major scale has one note that breaks the T T S T T T S pattern. Which, and what should it be?

Answer: 1 F should be F♯ · 2 C should be C♯ · 3 B should be B♭.

PART III · SCALES AND KEYS · 1 · CONTINUED

TRY IT · 10 SECONDS
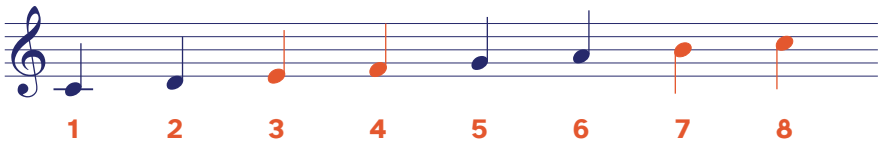


One finger, C to C.

Play the white keys from C up to C and say “tone, tone, semitone, tone, tone, tone, semitone” as you go. Feel the two short steps.

EXAMPLE 3

Where are the semitones?
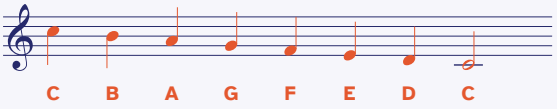


1. **Number the degrees** 1 to 8 from C.
2. **Look for no black key between:** E–F and B–C.
3. **Those are degrees 3–4 and 7–8.**

Every major scale has its semitones between **3–4** and **7–8**.

YOU KNOW THIS

Joy to the World



The first line is the **C major scale**, top to bottom: C B A G F E D C.

Try it. Play C major down from the top C and sing “Joy to the world” over it.

THINK OF IT A scale is a staircase with two short steps. Wherever you build it, the short steps sit in the same places: between 3 and 4, and between 7 and 8.

TRAP

- × Using the same letter twice: every letter appears once (F♯, not G♭, in G major).
- × Forgetting the semitone between 7 and 8: the leading note pulls up to the key note.

SEE ALSO

Minor scales p. 73 · Keys and key signatures p. 76

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Scales

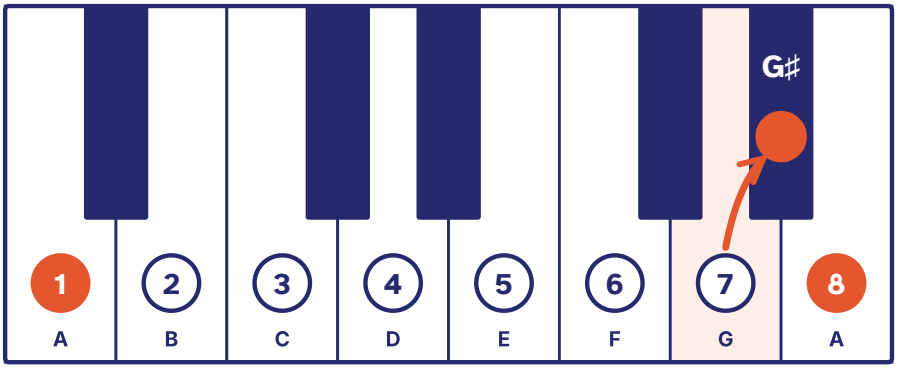
PART III · SCALES AND KEYS · 2

Minor scales

Every minor scale starts from the natural minor, the notes of its relative major from the 6th degree up. Two small changes give the harmonic and melodic forms, which sound stronger at the top.
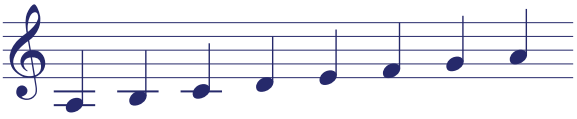
THE SECRET

Three minor scales, one home note: only the 6th and 7th change.
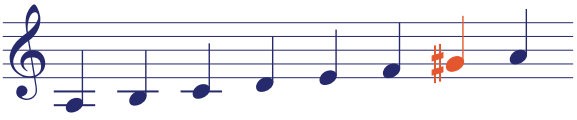


From A on the white keys · then raise the 7th: G♯

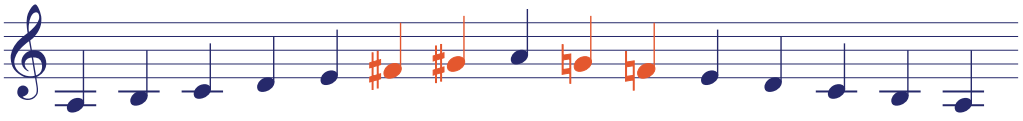
A minor in three forms



Natural: the notes of C major, from A to A. T S T T S T T.



Harmonic: the 7th raised a semitone (**G♯**). The gap F–G♯ is a tone and a half.



Melodic: the 6th and 7th raised going up (**F♯ G♯**), lowered coming down (**G♮ F♮**).

Relative minor

Same key signature, three semitones below the major: C major and A minor share every note.

Parallel minor

Same key note, different signature: C major (no sharps or flats) and C minor (three flats).

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 1

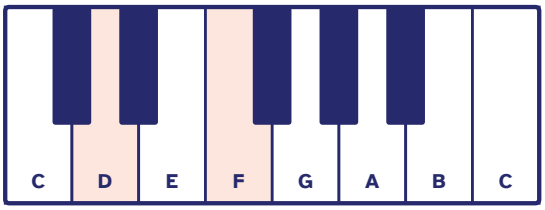
Build E harmonic minor

1. **Borrow G major's notes** starting on E: E F♯ G A B C D E.
2. **Raise the 7th:** D becomes D♯.
3. **Check the gap** C to D♯: a tone and a half.

E harmonic minor: **E F♯ G A B C D♯ E**.

EXAMPLE 2

Find the relative minor of F major



1. **Start on F,** the key note.
2. **Count three semitones down:** E, E♭, D.
3. **Check the signature:** D minor has one flat, B♭, like F major.

The relative minor of F major is **D minor**.

GAME

Which minor?

1	2	3
nothing raised	the 7th raised	6th and 7th raised going up, lowered coming down

Three forms of the minor scale. Name each one.

Answer: 1 natural · 2 harmonic · 3 melodic.

TRY IT · 10 SECONDS
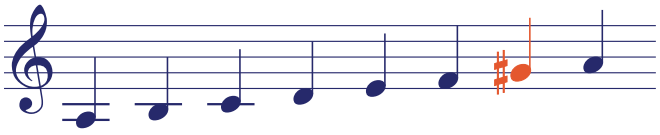


A to A on the white keys.

That is A natural minor: dark from the first note. Play it again with G♯ instead of G: harmonic minor, and the top pulls home.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 3

Natural or harmonic minor?

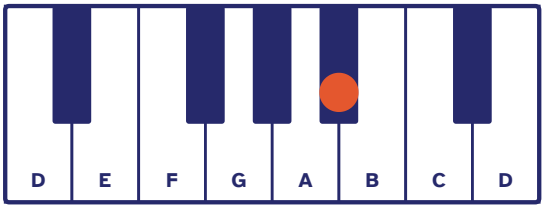


1. **Start on A** and read up to A.
2. **Look at the 7th:** G♯, raised.
3. **Nothing else changes** from the natural minor.

A raised 7th only: the **harmonic minor**.

EXAMPLE 4

Build D natural minor
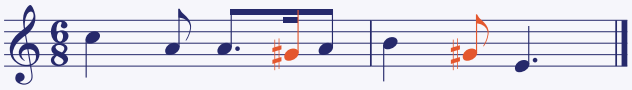


1. **Find its relative major:** three semitones up from D is F.
2. **F major has one flat:** B♭.
3. **Play F major's notes from D to D:** D E F G A B♭ C D.

D natural minor: **D E F G A B♭ C D**, with the same key signature as F major.

YOU KNOW THIS

Greensleeves



A minor, with a G♯: the raised 7th of the harmonic minor, leaning hard up to A.

Try it. Play it once with G♯, once with G: hear the pull disappear.

THINK OF IT The relative minor is the same clothes worn by a different person: the same notes, a new key note. The parallel minor is the same person in different clothes: the same key note, some notes changed.

TRAP

- ✕ Raising the 7th in the key signature: it is written as an accidental each time.
- ✕ Playing the melodic minor's raised notes on the way down.

SEE ALSO

Major scales p. 70 · Keys and key signatures p. 76

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Scales

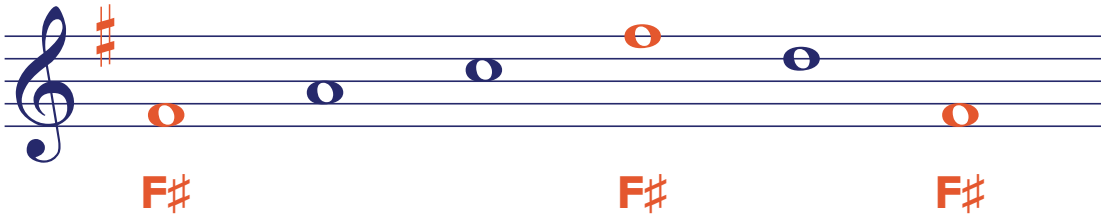
PART III · SCALES AND KEYS · 3

Keys and key signatures

Why write F♯ fifty times? The **key signature** at the start of each line says it once, for the whole piece. Its sharps or flats always line up in the same order.

THE SECRET

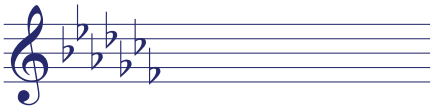
The key signature says once, at the start, which notes are always sharp or flat.



One sharp at the start: every F plays F♯



Sharps: F C G D A E B. The key note is a semitone above the last sharp.



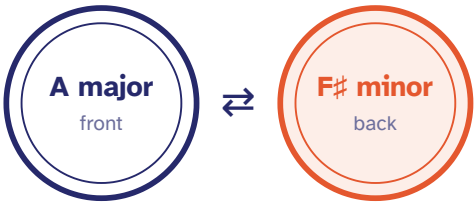
Flats: B E A D G C F. From two flats, the second-last flat names the key.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

MAJOR	RELATIVE MINOR	SIGNATURE	FIRST MET
C major	A minor	—	Level ½
G major	E minor	1♯	Level ½
D major	B minor	2♯	Level 1
A major	F♯ minor	3♯	Level 3
E major	C♯ minor	4♯	Level 4
B major	G♯ minor	5♯	Level 4
F major	D minor	1♭	Level ½
B♭ major	G minor	2♭	Level 1
E♭ major	C minor	3♭	Level 3
A♭ major	F minor	4♭	Level 4
D♭ major	B♭ minor	5♭	Level 4

Six and seven sharps or flats (F♯, C♯, G♭, C♭ major and their minors) come at Levels 4–6.

PART III · SCALES AND KEYS · 3 · CONTINUED

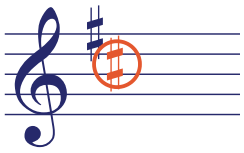


One signature, two keys

Every key signature belongs to **one major key and one minor key**, like the two sides of a coin: three sharps are A major or F♯ minor. The minor is the **relative minor**, three semitones below the major.

EXAMPLE 1

Name the keys



1. **Count:** two sharps, F♯ and C♯.
2. **The last sharp** is C♯; one semitone up is D.
3. **The relative minor,** three semitones below D, is B.

Two sharps: **D major** or **B minor**.

GAME

How many sharps or flats?

1

G major

2

D major

3

A major

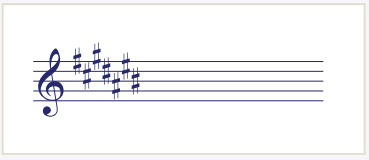
4

F major

Use the table on p. 76: how many sharps or flats does each key have?

Answer: 1 one sharp · 2 two · 3 three · 4 one flat.

TRY IT · 10 SECONDS
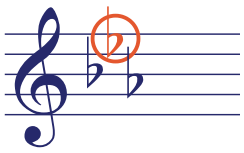


Say the sharps in order.

"Father Charles Goes Down And Ends Battle": F C G D A E B. Then backwards for the flats: B E A D G C F.

EXAMPLE 2

Read a flat key signature



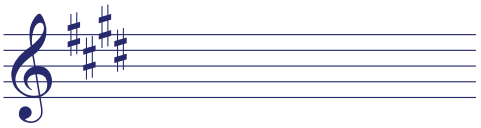
1. **Count the flats:** three, B♭ E♭ A♭.
2. **The second-last flat** is E♭: the key note.
3. **Its relative minor,** three semitones down, is C minor.

Three flats: **E♭ major** or **C minor**.

PART III · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 3

Read four sharps



1. **Name the sharps:** F♯ C♯ G♯ D♯.
2. **The last sharp is D♯;** one key up is E.
3. **Two keys share it:** E major, and its relative minor, C♯ minor.

Four sharps: **E major** or **C♯ minor**.

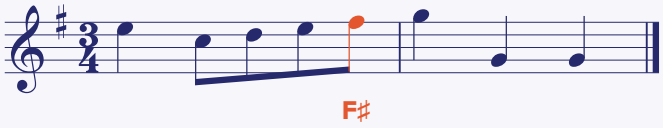
HOW TO

Write a key signature

1. **Sharps** always come in the same order, F C G D A E B, each on its own line or space.
2. **Flats** come in the reverse order, B E A D G C F.
3. **Write it again** at the start of every line, in both staves.

YOU KNOW THIS

Minuet in G



One sharp in the key signature: every F is F♯, like the one just before the top G. That sharp is why it is in G major.

Try it. Find every F in the piece and play it as F♯.

THINK OF IT A dress code on a party invitation: written once at the top, it applies to every guest all night. Only an accidental, a word to one guest, overrules it, and only until the bar line.

TRAP

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Forgetting that a signature names two keys: a major and its relative minor.
- × Reading it once and forgetting it: glance at the key signature at the start of every line.

SEE ALSO

Major scales p. 70 · Finding the key p. 79 · The circle of fifths p. 82

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Keys

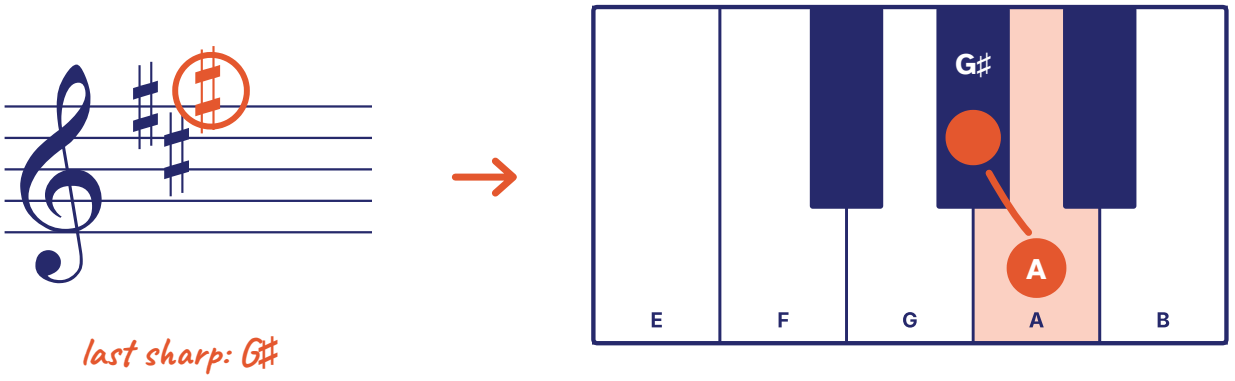
PART III · SCALES AND KEYS · 4

Finding the key

A key signature names two keys, a major and its relative minor. Two quick rules (and one to remember) find the major; one more finds the minor; then the music itself tells you which of the two it is.
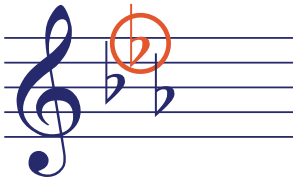
THE SECRET

Last sharp, one key up: the major key. Last flat but one: the major key.



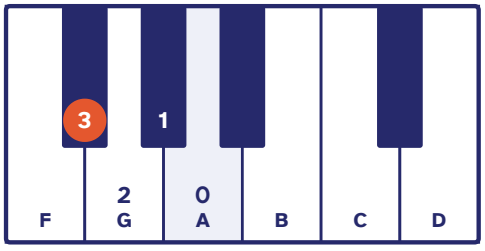
last sharp: G♯

Last sharp, one key up: A major



Flats: the second-last flat is the key.

B♭ **E♭** A♭ → **E♭ major**. One flat alone is F major.



Major to minor: three semitones down.

A, G♯, G, **F♯** → **F♯ minor**, the relative minor of A major (count every key, black and white).

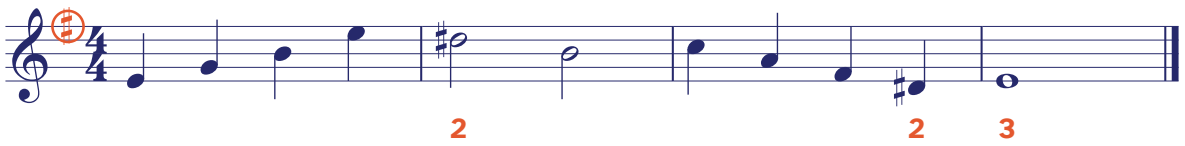
Minor keys and their clue notes

MINOR KEY	SIGNATURE	CLUE NOTE (RAISED 7TH)
A minor	none	**G♯**
E minor	1♯	**D♯**
D minor	1♭	**C♯**
B minor	2♯	**A♯**
G minor	2♭	**F♯**

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 1

What key is this tune in?

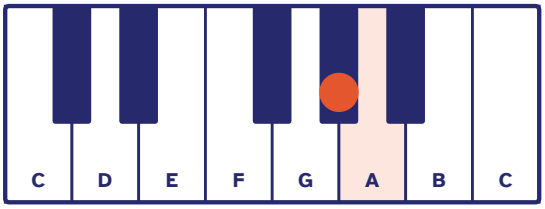


1. **The key signature:** one sharp, so G major or its relative, E minor.
2. **The clue notes:** D♯ keeps appearing. D♯ is the raised 7th of E minor; G major has no D♯.
3. **The last note:** E, the key note of E minor.

The key: **E minor**. Check it the same way with any piece you play.

EXAMPLE 2

Major or minor? No sharps or flats



1. **The signature:** none, so C major or A minor.
2. **The clue notes:** G♯ appears. G♯ is the raised 7th of A minor.
3. **The last note:** A.

No sharps or flats, G♯ clues, ending on A: **A minor**.

GAME

Name the major key

1	2	3	4
2 sharps	4 sharps	1 flat	3 flats

Sharps: the last sharp, one key up. Flats: the second-last flat is the key (one flat alone is F).

Answer: 1 D · 2 E · 3 F · 4 E♭.

TRY IT · 10 SECONDS



Three sharps, stop, step up.

Say “F C G”, stop: the last one is G♯. One key up is A. Three sharps is A major. Try it with two and four.

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 3

Read two flats



1. **Name the flats:** B♭ and E♭.
2. **The second-last flat** is B♭.
3. **Its relative minor** is three semitones below: G.

Two flats: **B♭ major** or **G minor**.

HOW TO

Find the key of a new piece

1. **The key signature:** name the major key and its relative minor.
2. **The clue notes:** look for the minor key's raised 7th.
3. **The last bass note:** pieces nearly always end on the key note.

EXAMPLE 4

Find a minor key from its clue



1. **Two flats:** B♭ major or G minor.
2. **G minor's 7th note is F;** raised, it becomes F♯.
3. **F♯s** through the piece and an ending on G mean G minor.

Two flats plus F♯s: **G minor**.

YOU KNOW THIS

Für Elise (Beethoven)



No key signature, a G♯ as the clue, and the music keeps coming home to A: **A minor**.

Try it. Play A, then G♯ A: hear the G♯ lean into home.

THINK OF IT Detective work: the key signature narrows it to two suspects; the clue note and the last note name the culprit.

SEE ALSO
Keys and key signatures p. 76 · The circle of fifths p. 82

PRACTISE IT
Workbook Scales and keys 1 · Online: Library › Keys

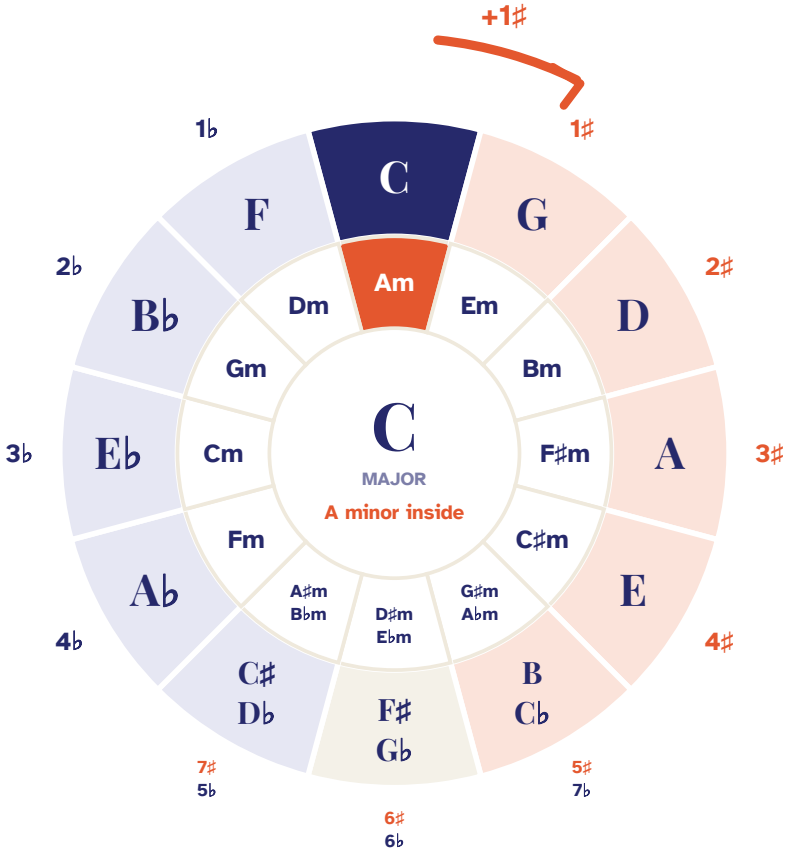
PART III · SCALES AND KEYS · 5

The circle of fifths

All the keys on one clock face. Going round clockwise, each key is a **5th higher** than the last and has **one more sharp** (or one fewer flat). Anticlockwise, each is a 4th higher, a 5th lower, with one more flat.

THE SECRET

Clockwise adds a sharp, anticlockwise adds a flat: every key on one clock.



Clockwise, one more sharp each step

How to read the clock

Outer ring: major keys. **Middle ring:** their relative minors (small letters). **Inner ring:** the key signature.

Neighbours are close

Keys next to each other share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor. These are the **related keys**.

PART III · SCALES AND KEYS · 5 · CONTINUED

Six jobs for one circle

Key signatures
Count the steps from C: clockwise for sharps, anticlockwise for flats.

Relative minors
Look straight in from any major key.

The main chords
A key's two neighbours give its other main chords: in C, the chords of F and G.

Cadences
The strongest close, V to I, is one step anticlockwise: G to C.

Changing key
Pieces usually move one step round, or to the relative minor.

Scale practice
Go round the circle: one new sharp or flat each time.

The first steps round the circle

STEPS	CLOCKWISE (SHARPS)	ANTICLOCKWISE (FLATS)
1	G major: F♯	F major: B♭
2	D major: F♯ C♯	B♭ major: B♭ E♭
3	A major: F♯ C♯ G♯	E♭ major: B♭ E♭ A♭

C major's neighbours

KEY	WHERE ON THE CIRCLE	WHY IT MATTERS
G major	one step clockwise	the dominant: the strongest pull home
F major	one step anticlockwise	the subdominant: a gentle move away
A minor	straight in from C	the relative minor: the same notes
E minor, D minor	in from G and F	the relatives of the neighbours

EXAMPLE 1

Why do G major and D major sound so close?

1. **Find both:** D is one step clockwise from G.
2. **Compare the scales:** G major has C; D major has C♯.
3. **Count:** six notes the same, one different.

Neighbours share six of their seven notes, so music moves between them easily.

EXAMPLE 2

Find A major's key signature

1. **Start at C** at the top of the circle: no sharps.
2. **Go clockwise:** G, D, A is three steps.
3. **Three steps,** three sharps: F♯, C♯, G♯.

A major has **three sharps**.

PART III · SCALES AND KEYS · 5 · CONTINUED

GAME

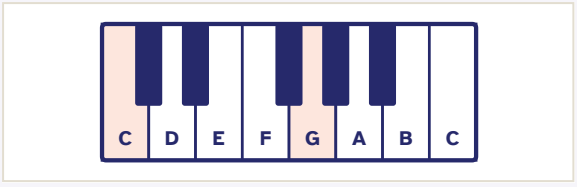
Neighbours on the clock

1	2	3	4
right of D	left of C	opposite C	inside G

Clockwise is right. Inside each major sits its relative minor.

Answer: 1 A · 2 F · 3 F♯ / G♭ · 4 E minor.

TRY IT · 10 SECONDS



Walk the circle: C G D A E.

Play each key note with one finger, a 5th up each time (drop an octave when you run out of hand). You just walked four steps clockwise.
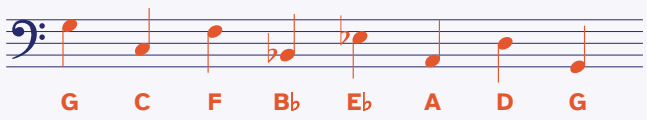
HOW TO

Learn the circle by heart

1. **Clockwise, out loud:** C G D A E B F♯. Each is a 5th above the last.
2. **Anticlockwise:** C F B♭ E♭ A♭ D♭ G♭. Each is a 5th below.
3. **Check yourself:** the number of steps from C is the number of sharps or flats.

YOU KNOW THIS

Passacaglia (Handel), the bass



Each bass note is a **5th below** the one before (or a 4th above): G C F B♭ E♭ A D G, one step round the circle each chord (except E♭ to A, a slightly smaller 5th that keeps the music in G minor).

Try it. Play the bass line slowly and hear it pull towards the last G.

THINK OF IT Like a colour wheel: neighbours blend, opposites clash. C and G sit side by side; C and F♯ sit opposite.

TRAP

- × Going the wrong way round: clockwise adds sharps, anticlockwise adds flats.
- × Forgetting the relative minors on the middle ring.

SEE ALSO

Keys and key signatures p. 76 · Finding the key p. 79 · Chords in a key p. 93

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Related keys

PART III · SCALES AND KEYS · TIPS

When a note sounds wrong

You read the note right, yet it sounds sour. The slip is usually in the key, not the note. Name the slip first, then fix only that note.
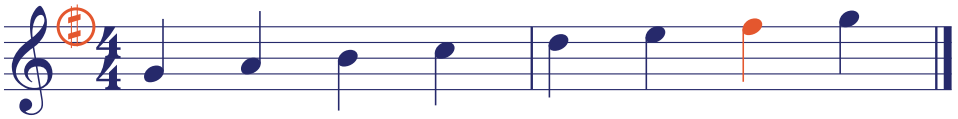
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A note sounds sour, though you read it right	**the key signature**: is this note sharp or flat all the way through?	p. 76
A key with many sharps or flats feels lost	**the circle**: clockwise adds a sharp, anticlockwise a flat	p. 82
A minor tune sounds wrong near the end	**the raised 7th**: harmonic and melodic minor sharpen it	p. 73
A scale sounds wrong in the middle	**the semitone steps**: 3–4 and 7–8 in every major scale	p. 70
A sharp is missing from the key signature	**the order of sharps**: F C G D A E B	p. 76
You are not sure which major key it is	**the last sharp**: the key is one note above it	p. 79

EXAMPLE

Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note sits on that line**, so the sharp belongs to it too.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

HOW TO

Three checks before you play

1. **The key signature:** say its sharps or flats aloud.
2. **The key:** name it from the last sharp or the last-but-one flat, then check the last note.
3. **The scale:** play it once, so your fingers know which black keys belong.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.